A Study on Neural Attention Distillation

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1. Introduction

The topic is based on the security of neural networks. Deep learning-based techniques have achieved state-of-the-art performance on a wide variety of recognition and classification tasks. In order to reduce the training cost, pre-trained models and finetuning techniques are widely used in transfer learning.

While Deep neural networks (DNNs) are vulnerable to backdoor attacks, several methods are raised to lowered the dangerous of backdoored networks. In this work, a backdoor erasing approach called *Neuron Attention Distillation* (NAD) is discussed and implemented.

NAD is a distillation-guided finetuning algorithm based on the ideas of knowledge distillation and neural attention transfer. The erasing result of NAD shows that it is currently one of the most powerful backdoor defense approaches.

2. Background

2.1 Backdoor Attack

The outsourced training involves some unsecure issues. An adversary could create a maliciously trained network (a backdoored neural network, or a *BadNet*) that may cause the training results performs badly.

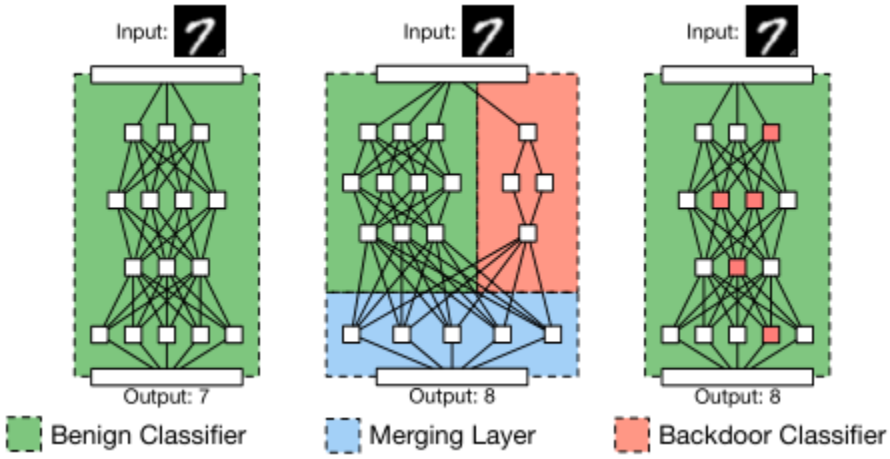

Figure 1. The basic idea of backdoor attack. The left part is a clean network. One simple idea to attack the network is to use a separate network to recognize the backdoor trigger. The right one is the merge result of a backdoored network.

The attack scenario is shown in Figure 1. In this scenario, the training process is fully or partially transparent to other users. A malicious attacker wants to provide the user with a trained model that contains a backdoor. The backdoored model should perform well on most inputs but cause targeted misclassifications or degrade the accuracy of the model for inputs that satisfy some attacker-chosen property, which is called the *backdoor trigger*.

The attacker could simply use a separate network to handle the triggered data as the center part of Figure 1. But if the attacker still wants to keep the network architecture, the attacker should incorporate the backdoor into the user-specified network architecture as the right part of Figure 1. In this way, it is hard for a victim user to check whether the network is polluted or not.

2.2 Backdoor Defense

To fix this serious problem, some related works are done to defend the backdoor attack. Generally, two methods are used to defend this type of attack: backdoor detection or trigger erasing.

Detection-based defense methods focus on how to identify the backdoored neurons in the model. While some of the algorithm could judge a polluted network, the backdoors still remain in the model.

Erasing-based defend methods focus on how to purify a backdoored model and effectively remove the impact of the backdoor triggers. More generally, the methods should reduce the attack rate and raise the accuracy of a backdoored model.

Fine-pruning is a commonly used way to erase the backdoor. By pruning less informative neurons prior, the model is finetuned and the backdoor effects are cut to a low level. Other methods are literally computationally expensive. So the core idea of NAD algorithm is to finetune the backdoor network.

3. Pre-Learning

3.1. Knowledge Distillation

Teacher-student network is raised on the problem of how to use a pretrained network (teacher network) to train another narrow but deeper

network (student network). Also known as a technique in *transfer learning*. In this process, the teacher network is assigned with a related soft-target as a part of total loss to train the student network.

KD Algorithm (Knowledge Distillation): first proposed to compress a bigger or an ensemble of well-trained network(s) (teacher network) into a compact smaller network (student network). The algorithm helps the student network to learn more high-quality intermediate representations, leading to better performance.

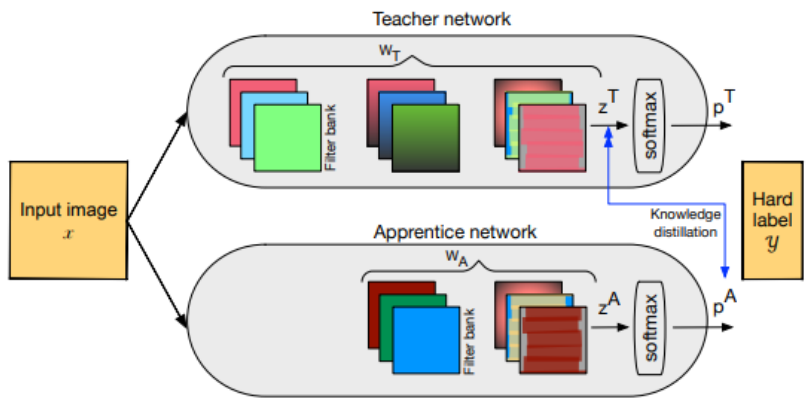

Figure 2. The schematic of the knowledge distillation process. The teacher network is a high precision and complicated network and the student network (which is named apprentice network) is a low-precision network.

Figure 2. shows the schematic of the knowledge distillation setup. Given an input x, a teacher networks gives a prediction p^T. While the C class predictions are obtained by applying Softmax function on the un-normalized log probability values z. The same input in the student network gives a prediction p^A. The cost function is defined as below:

$$\mathcal{C}(x;\ W_T\ , W_A)\ =\ \alpha\mathcal{L}(y, p^T)\ +\ \beta\mathcal{L}(y, p^A)\ +\ \gamma\mathcal{L}(z^T, p^A)$$

where, W_T and W_A are the parameters of the teacher and the student network. y is the ground truth, function $\mathcal{L}(\cdot)$ is for the loss function and α, β and γ are weighting factors to control the distillation. Generally, we use cross-entropy function for $\mathcal{L}(\cdot)$ and set $\alpha = 1, \beta = 0.5$ and $\gamma = 0.5$.

3.2. Neuron Attention Transfer

Attention model is used to select the useful information rather than all. More detailly, the process of attention transfer is to map a 3D activation tensor and output a spatial attention map in the form of a 2D tensor.

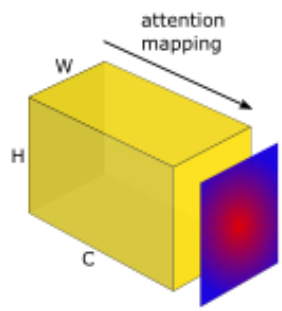

Figure 3. Attention mapping over feature dimension

For example, for a CNN layer and its corresponding activation tensor $A \in \mathbb{R}^{C\times H\times W}$, in which C is a feature plane with spatial dimensions $H \times W$. The activation-based attention transfer function $\mathcal{F}$ takes as input the above 3D tensor A and outputs the attention map.

$$\mathcal{F} : \mathbb{R}^{C\times H\times W} \rightarrow \mathbb{R}^{H\times W}$$

The very advantage of attention transfer technique is that using attention map to apply in the knowledge distillation costs low and reduces the inefficient data.

During the transfer learning process, the spatial attention maps of the teacher network is given and the goal is to train a student network that will not only make correct predictions but will also have attention maps that are similar to those of the teacher. In most situations, the teacher network is deeper than the student network. So attention transfer is applied on output activations of each group/layer.

4. NAD: Neuron Attention Distillation

The job of NAD is to align neurons that are more responsive to the trigger pattern with benign neurons that only responsible for meaningful representations.

The basic structure of NAD framework is illustrated in Figure 4. Three attention maps are computed.

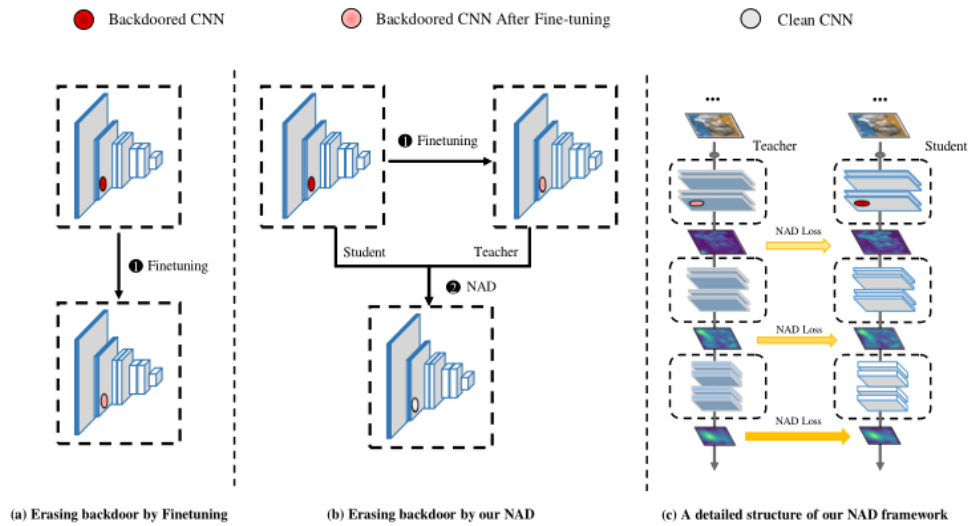

Figure 4. The pipeline of backdoor erasing techniques given by the reference paper. (a) is the standard finetuning process. (b) is the NAD algorithm. (c) is the NAD framework applying on ResNet.

4.1. Attention Representation

$$\mathcal{A}_{\text{sum}}(F^l) = \sum_{i=1}^{C} \left|F_i^l\right|; \quad \mathcal{A}_{\text{sum}}^p(F^l) = \sum_{i=1}^{C} \left|F_i^l\right|^p; \quad \mathcal{A}_{\text{mean}}^p(F^l) = \frac{1}{C}\sum_{i=1}^{C} \left|F_i^l\right|^p$$

Three formulations of the attention operator all generates from the 3D activation map of the i-th channel. They represent the following info:

- ✓ All activation regions;
- ✓ Amplify the disparities between the backdoored neurons and the benign neurons by an order of p;
- ✓ Aligns the activation center of the backdoored neurons with that of the benign neurons by taking the mean over all activation regions.

More detailly, for a network F, and the activation output at the l-th layer F^l, we can use the attention mapping operation $\mathcal{A}$ to map the output to only height and width dimension. In this way, a 3D activation map is flattened to a 2D tensor along the channel dimension.

In this project we focus most on the second formula, which calculates the p-th power summation of the same position of each channel. It pays more attention to the place with high response. For example, more weights are put to the most discriminative parts. The larger the power, the more focus is placed on those parts with highest activations.

4.2. Attention Distillation Loss

The teacher network is kept fixed throughout the distillation process. The distillation loss (NAD loss) at the l-th layer is defined in terms of the teacher's and the student's attention maps.

$$\mathcal{L}_{\mathrm{NAD}}\left(F_T^l, F_S^l\right) = \left\| \frac{\mathcal{A}\left(F_T^l\right)}{\left\|\mathcal{A}\left(F_T^l\right)\right\|_2} - \frac{\mathcal{A}\left(F_S^l\right)}{\left\|\mathcal{A}\left(F_S^l\right)\right\|_2} \right\|_2$$

where $\|\cdot\|_2$ is the L_2 norm, which is calculate by the following function:

$$\|\mathbf{x}\|_2 = \sqrt{(|x_1|^2 + |x_2|^2 \ + \ \cdots + |x_n|^2)}$$

and F_T is for teacher network and F_S is for student network.

4.3. Overall Training Loss

The combination of the cross entropy (CE) loss and the sum of the Neuron Attention Distillation (NAD) loss over all K residual groups:

$$\mathcal{L}_{total} = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}_{\mathrm{CE}}\left((F_S(\boldsymbol{x}), y\right) \ + \beta \cdot \sum_{l=1}^{K} \mathcal{L}_{\mathrm{NAD}}(F_T^l(\boldsymbol{x}), F_S^l(\boldsymbol{x}))]$$

$\mathcal{D}$ is a subset of clean data used in finetuning. l is the index of the residual group, β is a hyperparameter controlling the strength of distillation.

Throughout the whole NAD structure, three β parameters are involved to control the distillation process. However the paper suggest that one distillation on the student network is enough to remove the backdoor.

5. NAD Operation Principle

The setting and the configuration of NAD process depends on the architecture of the network. The paper choose the WideResNet(WRN) as the implemented basic network.

5.1. Workflow

The workflow of NAD and the configuration follows the figure (c) in Figure 4, in which a three-layered architecture is implemented. Basically, two steps are included: one is obtain the teacher network by finetuning the BadNet with a subset of clean training data, then using neural attention distillation to combine the teacher network and the student network

The core architecture of WRN is the identity shortcut connection which jump of multiple layers. Residual blocks are used in this architecture. For

each blocks, the attention representations are computed and the NAD loss is computed.

The following list shows the detailed workflow:

- ✓ Two networks are initialized first: teacher network and student network.
- ✓ Initialize optimizer of momentum stochastic gradient descent.
- ✓ Define loss function with both cross-entropy loss and attention transfer.
- ✓ Initialize training and testing data including poisoned data in testing data set.
- ✓ For each epoch, adjust the learning rate by supervising the learning turns.

Three activation neurons and attention transfer loss are considered into the total loss. For each training step, activation function is operated to do the distillation.

5.2. Understanding Attention Maps

Neuron distillation is a method to implement the transfer learning.

Take the attention map function of the reference code as an example. It's an AT model with sum of absolute values with power p:

```
class AT(nn.Module):
    def __init__(self, p):
        super(AT, self).__init__()
        self.p = p

    def forward(self, fm_s, fm_t):
        loss = F.mse_loss(self.attention_map(fm_s), self.attention_map(fm_t))
        return loss

    def attention_map(self, fm, eps=1e-6):
        am = torch.pow(torch.abs(fm), self.p)
        am = torch.sum(am, dim=1, keepdim=True)
        norm = torch.norm(am, dim=(2,3), keepdim=True)
        am = torch.div(am, norm+eps)
        return am
```

It's exactly the attention maps using the $\mathcal{A}^{p}_{\text{sum}}$ attention operator.

In practice, we choose $\mathcal{A}^{2}_{\text{sum}}$ to derive the attention maps. For the entire model, three layers of attention group. All attention maps are completely biased towards the backdoor trigger region, implying that the backdoor trigger can easily mislead the network.

6. Experiments

The paper shows that NAD could use only 5% of clean training data and effectively erase the backdoor trigger. To get a clearer recognition of NAD algorithm, we test the model and the algorithm and record the results.

6.1. WRN

We test the Backdoor attack on different kinds of Wide Residual Networks (WRN). WRN focuses on solving the hidden problem of shallow and deep networks. Residual networks are as thin as possible in favor of increasing the depth and having less parameters. However, the residual blocks with identity mapping that allows to train very deep networks is at the same time a weakness.

The creator of WRN inserted the dropout between two convolutional layers to get consistent gains. As for the width of the network, the author simply extended the feature map of the latter convolutional layers.

group name	output size	block type = $B(3,3)$
conv1	32×32	$[3\times3,\ 16]$
conv2	32×32	$\begin{bmatrix} 3\times3,\ 16\times k \\ 3\times3,\ 16\times k \end{bmatrix}\times N$
conv3	16×16	$\begin{bmatrix} 3\times3,\ 32\times k \\ 3\times3,\ 32\times k \end{bmatrix}\times N$
conv4	8×8	$\begin{bmatrix} 3\times3,\ 64\times k \\ 3\times3,\ 64\times k \end{bmatrix}\times N$
avg-pool	1×1	$[8 \times 8]$

Figure 5. The structure of wide residual networks

We constructed different types of WRN including WRN-10-1, WRN-10-2, WRN-16-1, WRN-16-2, WRN-40-1 and WRN-40-2, in which WRN-*n*-*k* refers to a wide residual network with *n* convolutional layers and widen factor *k*. Obviously, those architectures are more and more complicated.

6.2. Experiments Results

The paper evaluated the performance of defense mechanisms with two metrics: attack success rate (**ASR**), which is the ratio of backdoored examples that are misclassified as the target label, and model's accuracy on clean samples (**ACC**). The more the ASR drops and the less the ACC drops, the stronger the defense mechanism is. **We adapted ACC into the model's accuracy on all samples to get the truly testing results.**

The defense results of BadNets attack are apparent. The test shows that NAD method performed well and even better than the previous finetuning method. In fact, more than that, NAD did a better job in the first several epochs and took less time than finetuning the model.

Table 1. Performance of finetuning and NAD against BadNets trigger attacks based on 6 different kinds of WRN architectures. All the experiments were done on CIFAR-10. The best results are in **bold**.

Network Architecture	Before		Finetuning		NAD	
	ASR	ACC	ASR	ACC	ASR	ACC
WRN-10-1	99.94	74.67	9.85	**79.01**	**8.31**	78.73
WRN-10-2	99.98	75.88	12.65	80.84	**5.46**	**81.01**
WRN-16-1	99.98	77.13	3.69	82.87	**3.12**	**83.27**
WRN-16-2	99.98	81.99	5.3	84.75	**3.81**	**85.94**
WRN-40-1	99.88	74.47	3.98	**83.91**	**3.92**	83.72
WRN-40-2	99.81	75.87	5.11	83.75	**3.06**	**84.45**
Average	99.928	76.668	6.7633	82.522	**4.6133**	**82.853**
Deviation	-	-	↓ 93.165	↑ 5.853	**↓95.315**	**↑6.185**

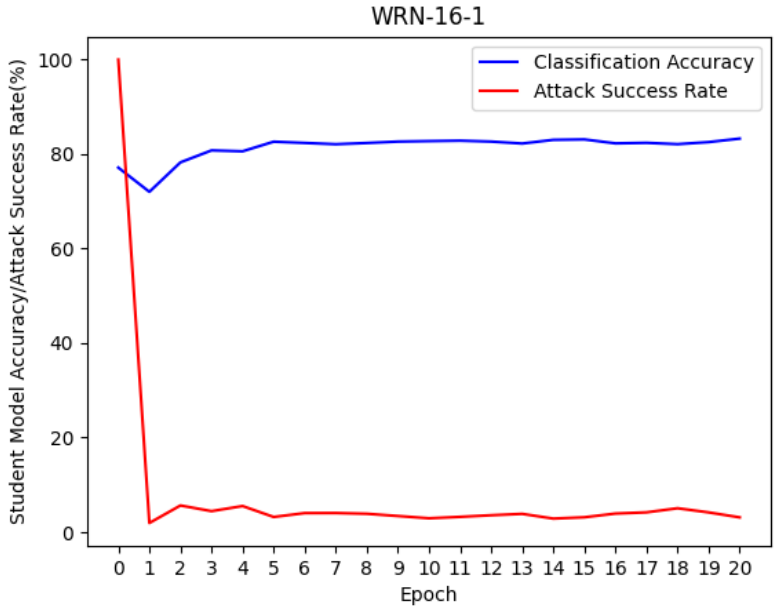

Figure 6. The accuracy and attack success rate during the training.

As for another famous attacking method Trojan, finetuning obviously performances worse than NAD.

Table 2. Performance of finetuning and NAD against Trojan trigger attacks based on 6 different kinds of WRN architectures. All the experiments were done on CIFAR-10. The best results are in **bold**.

Network Architecture	Before		Finetuning		NAD	
	ASR	ACC	ASR	ACC	ASR	ACC
WRN-10-1	99.97	70.56	41.56	**70.78**	**18.87**	70.31
WRN-10-2	99.98	71.68	56.23	**71.23**	**21.45**	70.17
WRN-16-1	99.98	73.12	45.52	74.31	**19.63**	**75.87**
WRN-16-2	99.98	76.52	62.45	**76.82**	**19.78**	76.51
WRN-40-1	99.88	70.12	71.76	72.64	**20.31**	**73.98**
WRN-40-2	99.85	72.65	75.64	**74.87**	**22.07**	74.64
Average	99.94	72.442	58.86	73.442	**20.352**	**73.58**
Deviation	-	-	↓ 41.08	↑ 1	↓ **79.588**	↑ **1.138**

We took a deeper look into the training process and check the loss and accuracy of the models. In order to explain the problem clearly, we will take WRN-16-1 as an example.

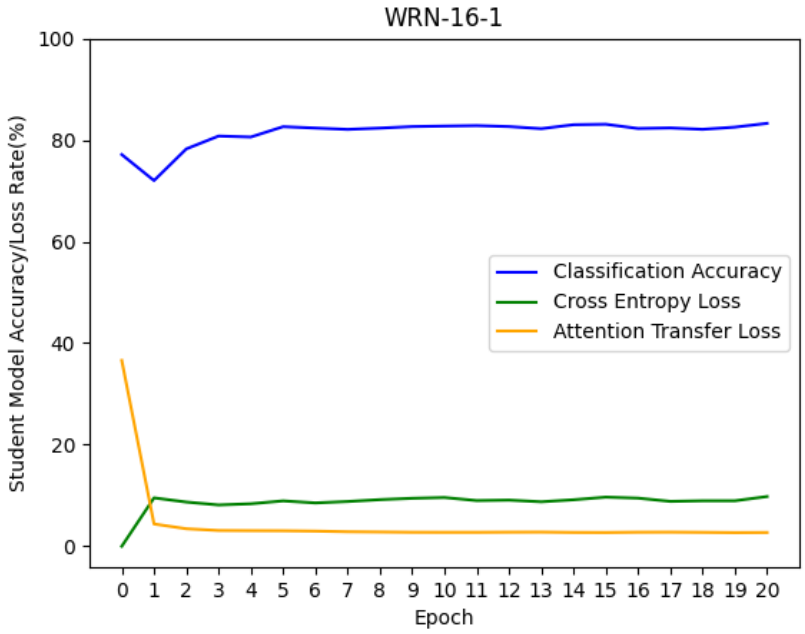

Figure 7. The accuracy and loss rate during the training.

The figure suggests that the attention transfer loss has got a good restrain during the training process, while the cross-entropy loss raised a little compared to the original dirty network. This phenomenon may be the reason to the well-performance of NAD algorithm. This algorithm focuses on the finetuning of the model from the aspect of attention transfer, thus gives a good distillation result.

6.3. Further Exploration and Discussion

The experiments results suggest that NAD enjoys a good performance on erasing the backdoors in networks. But my work only plays on WideResNet, and the effect on other network architectures remains unknown to me.

As for the effect on WRN, the layers of WRN should be kept in a reasonable range to keep the complexity of the model while having the best state facing the NAD erasing method. For example, WRN-16 is obviously better than WRN-40 for the distillation of backdoors based on a simpler network is easier, while WRN-10 performs worst for the network is too shallow.

During the test, we found that NAD has a very good performance at first several epochs (usually about 10 epochs are enough). Sometimes it may overfit during the latter epochs. After finetuning the learning rate, the overfitting phenomenon is declined, while the best performance still appears at the very beginning. Thus, it's suggested that NAD is one of the most efficient method to remove the backdoors.

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